

A quantum mechanical interaction of human erythrocytes¹

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Theory predicts that the membrane potential will polarize membrane molecules and cause them to vibrate coherently at a frequency of $\sim 10^{11}$ Hz. If the supply of metabolic energy exceeds a minimum value, membrane phonons may condense their momentum into a single "giant" vibrational mode. At 10^{11} Hz ionic screening is small up to distances of approximately a micrometre, so forces of a range several orders of magnitude longer than chemical forces can arise. These forces may be attractive or repulsive depending on frequency. They should occur in every metabolically active membrane and may control macromolecular transport and enzyme–substrate interactions. We find that normal human erythrocytes in plasma form rouleaux faster than Brownian motion predicts. When cells are fixed in glutaraldehyde or are metabolically depleted, or if the membrane potential is brought to zero, the rate of aggregation agrees with Brownian theory. When the metabolically depleted cells are revived or if the membrane potential is restored, then the interaction returns.

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La théorie veut que le potentiel de membrane polarise les molécules membraneuses et les fait vibrer cohéremment à une fréquence d'environ 10^{11} Hz. Si l'apport d'énergie métabolique dépasse une valeur minimale, les phonons membraneux peuvent concentrer leur momentum en un seul mode de vibration. A 10^{11} Hz, l'interaction ionique est faible jusqu'à des distances de quelques micromètres, de sorte que des forces de plusieurs ordres de grandeur plus élevées que les forces chimiques peuvent se développer. Ces forces peuvent être attractives ou répulsives suivant la fréquence. Elles devraient apparaître dans chaque membrane métaboliquement active et peuvent contrôler le transport macromoléculaire et les interactions enzyme–substrat. Nous constatons que les érythrocytes humains normaux contenus dans le plasma forment des rouleaux plus rapidement que ne le prévoit le mouvement brownien. Quand les cellules sont fixées dans le glutaraldéhyde ou appauvries métaboliquement ou encore si le potentiel de membrane est placé à zéro, le taux d'aggrégation concorde avec la théorie brownienne. Quand les cellules appauvries métaboliquement sont revivifiées ou lorsque le potentiel de membrane est rétabli, alors l'interaction se normalise.

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Introduction

For over a decade Fröhlich (1968) has been developing the concept of a unique collective interaction of the molecules comprising cell membranes. Collective interactions are well known in the physics of the condensed state of matter and are responsible for such subtleties as super-conductivity and the bizarre behaviour of liquid helium II. The manifestations of Fröhlich's theory should arise at normal temperatures and should occur in all living systems. In particular the

theory predicts the existence of very long-range forces ($\sim 1 \mu\text{m}$) which are highly specific in that an organelle or a cell could recognize another at a distance and be attracted or repelled. Fröhlich (1980) recently reviewed the theory and the evidence in its support.

Since the membrane potential acts across a membrane thickness of the order of 10 nm, the electric field is very great ($\sim 10^7 \text{ V} \cdot \text{m}^{-1}$), and membrane molecules will be strongly polarized in this field. Thermal vibration of the dipoles in the membrane correspond to a polarization wave of half wavelength roughly equal to the membrane thickness. If the velocity of mechanical waves in the membrane has the reasonable value of $10^3 \text{ m} \cdot \text{s}^{-1}$, the vibrations will have a frequency of $\sim 10^{11}$ Hz. Resonant absorption in tissue cultures and in microorganisms in the neighbourhood of this frequency comprises the main evidence for the theory (Fröhlich 1980). Since polar molecules interact electrically, the vibrations of the membrane constituents will be col-

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lective and the quantum-mechanical properties of these collective interactions will be apparent and important at 10^{11} Hz. The name phonon is given to this aspect of the vibration by analogy with the photon, the corresponding "quasi-particle" of the electromagnetic waves. In a strong electric field, conformational change in macromolecules and the readjustment of counter ions can result in minima appearing in the relationship between polarization and electric field. If these minima are above the energy of the ground state, they will correspond with metastable polar states; if lower than the ground state, the system will be ferroelectric. Even if the metastable states have a short life their occupancy may remain high by the exchange of thermal phonons with the surroundings, which is a heat bath in the thermodynamic sense. If energy, say from cell metabolism, is now supplied to the modes of such a system of collective polar excitations, a new process arises with the exchange of the difference in energy of two quanta with the heat bath. If this process dominates single-phonon exchange, the supply of energy will become concentrated into the mode of highest wave number and constitute a giant vibration. Phonons have zero spin and obey Bose-Einstein statistics. The mathematical formulation of the accumulation of energy into a single zero wave-number mode is similar to that of the Einstein condensation of a Bose gas (London 1950; Kittel 1958), a condensation in momentum space, not in coordinate space which is the case for the condensation of a common gas. Liquid helium II is presumed to result from such a quantum condensation (London 1950).

Fröhlich's theory as it stands is quite general and makes no predictions about specific cell systems. The important conclusion is that forces, of a range several orders of magnitude longer than London-van der Waals' forces should exist between macromolecular systems within the cell and between cells, perhaps over distances of micrometres. The forces may be attractive or repulsive depending on the natural frequencies of the interacting systems and the dispersion of the dielectric constant, as a function of frequency, of the medium in which the systems are suspended. The necessary features of an entity which might be expected to manifest a Fröhlich effect are (a) an organization of molecules into some system, (b) a membrane potential to polarize the array of molecules in the system, and (c) a supply of metabolic energy. In the course of experimentation on red-cell rouleau formation (Rowlands and Skibo 1972; Sewchand and Canham 1979) we noticed a small attraction and a "jiggling" of the membranes just prior to the contact which precedes doublet formation. When we upset the organization of the membrane or remove the supply of energy or alter the membrane potential, this attraction is affected. An account of our early results has been published (Rowlands et al. 1981).

Methods

Venous blood was drawn on the morning of an experiment and immediately mixed with heparin (14 USP units/mL of blood). The plasma was separated by centrifugation and filtered through a 0.3- μ m Millipore filter. Red cells, normal or experimentally modified, were added to the plasma at a concentration (v/v) of 0.5%. The low haematocrit blood was stirred rapidly with a magnetic stirrer to break up rouleaux and quickly transferred to a siliconized (Dow Corning, Midland, MI, Sylgard 184 or R4-3117) haemocytometer chamber (Spencer Bright-Line, American Optical Corporation, Buffalo, NY) using a 30-gauge needle to ensure high shear rate and so minimize prior rouleau formation. The chamber was sealed with silicone grease to prevent evaporation and photographed with a time-lapse cinécamera on an Ortholux microscope (Wild-Leitz, Rockleigh, NJ) at about 60 $\times$ magnification. An exposure was taken every 15 s for 3 h. About 150–200 cells were in view.

Glutaraldehyde-fixed cells

Red cells were washed twice in phosphate-buffered saline (PBS) containing 0.04% glutaraldehyde (PBS composition: 150 mM NaCl, 7.3 mM Na_2HPO_4 , 2.8 mM NaH_2PO_4 , 2.2 mM CaCl_2 , 4.6 mM MgCl_2 , 310 mosmol, pH 7.4 at 22°C), suspended in the same solution, and shaken gently for 24 h at 22°C. They were stored at 4°C until use. Cells so treated retain normal shape, area, and volume (R. E. Lovlin, unpublished observations).

Microspheres

Latex microspheres 3.4 μ m in diameter (Coulter Electronics, Hialeah, FL) were added to plasma at a concentration (v/v) of about 0.2%, shaken, and used immediately. Polystyrene DVB microspheres 5.9 and 8.0 μ m in diameter (Duke Scientific, Palo Alto, CA) stuck to the floor of the haemocytometer chamber when in plasma. They were therefore suspended in soapy water at a similar concentration.

Variation of membrane potential

The pH of the cell suspension was measured (PHM 64 Radiometer, Copenhagen, electrode GK2421C) and then lowered by adding 0.14 M HCl in 0.14 M NaCl (so keeping $[\text{Na}^+]$ constant) or increased by adding 0.14 M NaOH (Funder and Wieth 1966; Lassen et al. 1980). The membrane potential was calculated from $[\text{Cl}_i^-]/[\text{Cl}_o^-]$ using the Nernst equation. $[\text{Cl}^-]$ was measured with a Corning chloride meter (model 920 M, Corning, Medfield, MA).

Abolition of membrane potential by ionophore A23187

Flatman and Lew (1980) used this ionophore to study magnesium buffering in the human red cell and showed that entry of magnesium depolarized the membrane. We used their method to achieve an approximately zero potential in the following way. The cells were washed twice in PBS containing 10 μ M A23187 and then were suspended in plasma to which had been added 10 mM MgCl_2 , 10 μ M EGTA, and 10 μ M A23187. The pretreatment of the cells with the ionophore was necessary because it is readily adsorbed by albumin (Sarkadi et al. 1976).

Metabolic depletion and repletion

Cells were washed in PBS, resuspended in PBS, and incu-

bated at 37°C for 22 h. They were repleted by adding 30 mM adenosine to the suspending medium and incubating for a further 3 h (Weed et al. 1969; Meiselman and Baker 1977).

Viscosity

The viscosity of plasma ($\eta \sim 1.7 \times 10^{-3}$ Pa·s) for each subject and of the soapy water ($\eta = 1.06 \times 10^{-3}$ Pa·s) for the microspheres was measured with a cone and plate viscometer (Synchro-Lectric, Brookfield, Stoughton, MA).

Analysis of results

The phenomenon is a coagulation of particles undergoing Brownian movement on which is superimposed a specific interaction at a distance. Smoluchowski (1917) developed a theory for the coagulation of the particles of colloidal suspensions. The theory is reviewed in a shortened form by Chandrasekhar (1943) and experimental tests are described by Swift and Friedlander (1964). It is assumed that the diffusion coefficient of the individual particles is given by the Stokes-Einstein relation

$$[1] \quad D = \frac{kT}{6\pi\eta r}$$

where k is Boltzmann's constant, T absolute temperature, η the viscosity of the suspending medium, and r the particle radius. In the theory, Smoluchowski incorporates the possibility of a zone of attraction, radius $Z > r$, around each particle, which ceases to exist if $Z = r$, and if $Z < r$ a repulsive force is implied. Chandrasekhar (1943, equation 461) gives an expression for the variation in the total number of particles as a function of time. We also adopt Swift and Friedlander's (1964) simplifying assumption that all particles have the same radius (and the implication that all zones of attraction have the same radius $Z \neq r$). If, then, N is the total number of particles (single and multiple) the kinetics of coagulation is given by

$$[2] \quad \frac{1}{N_t} - \frac{1}{N_0} = \frac{4kTZ}{3\eta r} t$$

Equation 2 is for a three-dimensional process and would apply if the cells had the same density as plasma and so were distributed randomly in depth as well as in the other two dimensions of the chamber. Cells sediment, however, so the process is compressed into two dimensions and the rate constant will be higher by a factor of 1.5. Equation 2 becomes

$$[3] \quad \frac{1}{N_t} - \frac{1}{N_0} = \frac{2kTZ}{\eta r} t$$

which predicts a linear relation between N_t^{-1} and t .

The developed cinéfilm was projected with a photo-optical data analyser (model 224A, L-W International, Woodland Hills, CA), and in the earliest frame at which the cells had settled, the total number of cells and rouleaux (N_0) was counted in an area defined by the haemocytometer grid. The count was repeated every fifth frame (75 s) initially, and at increasing intervals up to 20 frames (300 s). The regression of N^{-1} on t was computed.

Einstein's (1926) theory of Brownian motion gives the mean square deviation in two dimensions as a function of time

$$[4] \quad \langle U^2 \rangle = 4kT\mu^{-1}$$

where μ is Stokes' drag coefficient for spheres

$$[5] \quad \mu = 6\pi\eta r$$

which is also the denominator in [1]. To check that isolated cells were executing Brownian motion, U was measured against t for each of 30 cells which did not come within 6 μ m of another during the first 15 min of each experiment. Beyond 15 min fewer than 30 cells fulfilled this criterion. The correlation of $\langle U^2 \rangle$ with t and the value of μ were calculated.

Results

Drag coefficients

The interaction occurs against a background of Brownian movement and, in differentiating between the two, the observations on the drag coefficient are pivotal. If the measured value of plasma viscosity is inserted into Stokes' Eq. 5, the drag coefficient for spheres of the same diameter as cells is 1.28×10^{-7} N·s·m⁻¹. Canham et al. (1971) observed the free fall of erythrocytes in Ringer's solution. Allowing for the different viscosity of plasma, we calculate from their published data $\mu = 1.49 \times 10^{-7}$ N·s·m⁻¹. The cells were tumbling as they fell so Perrin's correction to Stokes' formula for oblate spheroids (Tanford 1961) is appropriate, bringing the theoretical value up to $\mu = 1.54 \times 10^{-7}$ N·s·m⁻¹. Deformability of cells could well account for the difference in the last two figures. These two values are substantially less than the experimental values (Table 1). The results for microspheres give an explanation. For the small microspheres the drag coefficient by Stokes' Eq. 5 is 0.57×10^{-7} N·s·m⁻¹ and the measured value 0.62×10^{-7} N·s·m⁻¹, 1.1 times higher. On the cinéfilm these particles are seen to be going in and out of focus, rising and falling continually and so spending a negligible fraction of time in contact with the floor of the chamber. The larger spheres have a specific gravity of $\rho = 1.05$ and are suspended in soapy water ($\rho = 1$). They remain in focus and are therefore close to the surface. For 5.9- μ m spheres, the Stokes' $\mu = 0.59 \times 10^{-7}$ N·s·m⁻¹ ($\eta = 1.06 \times 10^{-3}$ Pa·s) whereas the experimental value is 1.8 times higher at 1.06×10^{-7} N·s·m⁻¹. Similarly for 8.0- μ m spheres the factor is 2.1. If we compare the mean μ for cells (Table 1) with the Stokes' μ for oblate spheroids of the same major diameter the factor is 1.78. We infer that the juxtaposition of either cells or large microspheres with the floor of the chamber introduces an additional hydrodynamic drag.

Since [2] and thus [3] are derived from [1] in which the denominator is the Stokes' coefficient, the rate constant in [3] has to be divided by the ratio of the experimentally determined to the theoretical drag coefficient (Table 1). In this way, for each experiment, a

TABLE 1. Drag coefficients

	$\mu \times 10^7 \text{ N} \cdot \text{s} \cdot \text{m}^{-1}$	Experimental value/ Stokes' value for oblate spheroids
Normal cells	2.80 ± 0.66^a	1.82 ± 0.43^a
Cells fixed in glutaraldehyde	2.96 ± 0.68	1.92 ± 0.44
Cells at pH 6.3	2.78 ± 0.76	1.80 ± 0.50
Cells restored to pH ~ 7.6	2.57 ± 0.83	1.67 ± 0.54
Ionophore-treated cells	3.39 ± 0.14	2.21 ± 0.09
Metabolically depleted cells	2.18 ± 0.34	1.42 ± 0.23
Cells repleted with adenosine	2.68 ± 0.74	1.74 ± 0.48
Mean value for cells	2.75 ± 0.10^b	1.78 ± 0.06^b
Microspheres 3.4 μm	0.62	1.1^d
Microspheres 5.9 μm	1.06^c	1.8^d
Microspheres 8.0 μm	1.67^c	2.1^d

^aStandard deviation.^bStandard error of the mean.^cThe larger microspheres in plasma stick to the floor of the chamber. These experiments were done in soapy water ($\eta = 1.06 \times 10^{-3} \text{ Pa} \cdot \text{s}$).^dThe Stokes' drag coefficient for spheres was used in this calculation.

"Brownian motion rate constant" (no interaction; $Z = r$) is calculated. The values are in the neighbourhood of $2.6 \times 10^{-18} \text{ m}^3 \cdot \text{s}^{-1}$. For convenience we use an "interaction coefficient" Ξ which has a magnitude of unity when representing the "Brownian motion rate constant". Slopes of the graph N_i^{-1} against t are reported in terms of Ξ and from [3]

$$[6] \quad \Xi = Z/r$$

To get a measurement of the accuracy of the determination of the drag coefficient, U ([4]) was measured in seven experiments on the same blood. The standard error of the mean value of $\langle U^2 \rangle$ was 19% which sets a lower limit to the accuracy of the experiments as presently performed. Correlation coefficients for the regression of $\langle U^2 \rangle$ on t averaged 0.94 and for N_i^{-1} on t averaged 0.98.

Normal cells, glutaraldehyde-fixed cells, and microspheres

In Table 2 the present results and those derived by an earlier method (Rowlands et al. 1981) are displayed together. The simple geometrical analysis, which we used previously, predicted that if there was no interaction the mean probability that pairs of cells, initially less than 8 μm apart, would collide in 3 h was 0.3. The theory used in analyzing the present results has ample experimental support (Swift and Friedlander 1964) from the coagulation of hydrosols. This theory predicts an attractive interaction if $\Xi > 1$. The figures in column

two, by having the same trend as those in column one, confirm our previous conclusions though the mathematical relationship between the two sets of figures is nonlinear, as might be expected.

The interaction coefficient Ξ for normal cells is three times greater than the theoretical value for neutral particles. The experimental value of $\Xi = 2.9$ taken together with a probability by the earlier method of 0.85 (value for no interaction ~ 0.3) is our evidence for an attraction at a distance between normal erythrocytes. Cells fixed in glutaraldehyde and the microspheres have significantly lower values for Ξ close to unity. Glutaraldehyde-fixed cells have a negative surface charge which is at least as large as that of normal cells (Vassar et al. 1972) but it will be neutralized by counter ions up to very short distances of approach ($\sim 1 \text{ nm}$) so an electrostatic repulsion is unlikely to be detected in the present experiments.

Variation in pH of the suspending medium

At pH 6.3 the membrane potential is close to zero (Funder and Wieth 1966), there should be no Fröhlich-type interaction, and Ξ should be equal to unity, which it is. At pH 6.3 the membrane potential calculated from the chloride ratio was 1.5 mV as against that in normal cells of 16 mV. Restoring pH to 7.6 raises Ξ significantly above unity and the membrane potential to 12 mV.

Ionophore-treated cells

Reducing the membrane potential by the procedure

TABLE 2. Interaction coefficients Ξ

	Probability ^a	Ξ^b	No. of experiments	No. of subjects
Normal cells	0.85	2.92 ± 1.12^c	15	9
Cells fixed in glutaraldehyde	0.46	0.92 ± 0.46	9	5
Cells at pH 6.3	0.40	1.07 ± 0.41	10	5
Cells restored to pH ~7.6	0.71	2.57 ± 0.99	9	3
Ionophore-treated cells	0.48	1.69 ± 0.04	2	2
Metabolically depleted cells	0.44	0.43 ± 0.11	4	2
Cells repleted with adenosine	0.87	3.26 ± 0.68	3	2
Microspheres 3.4 μm	0.24	0.72	1	—
Microspheres 5.9 μm	—	0.93	1	—
Microspheres 8.0 μm	—	1.15	1	—

^aProbability of pairs of cells colliding in 3 h (Rowlands et al. 1981). Value for no interaction ~0.3.

^bInteraction coefficient as defined in text. $\Xi > 1$ attraction.

^cStandard deviation: $p < 0.001$ normal versus glutaraldehyde cells; normal versus pH 6.3 cells; pH 6.3 versus pH 7.6 cells; normal versus metabolically depleted cells; $p < 0.003$ normal versus ionophore-treated cells.

of Flatman and Lew (1980) also reduced Ξ significantly. There were only two experiments so further work is needed to discover if the difference between the two methods of changing membrane potential is real. There are some difficulties common to both the ionophore and the metabolic depletion experiments which will be explained in the Discussion.

Metabolic depletion and repletion

Depleting the cells of ATP caused the largest decrease in Ξ into a range suggestive of a repulsion. Repletion with adenosine reversed the alteration completely. Metabolic depletion reduced the calculated membrane potential from 15.0 to 7.1 mV so the marked change might be the result of a combination of two effects which may be difficult to separate (see Discussion).

Range of the attraction

From [3] and the definition of Ξ , it follows that the range of attraction $Z = \Xi r$ where r is the radius of the cell. For normal cells $\Xi = 3$ and therefore $Z = 12 \mu\text{m}$. The implication is that cells separated beyond this zone execute Brownian movement. On closer approach, pairs of cells are, on the average, committed to make contact and to form a rouleau. Energetic fluctuations of Brownian movement can of course drive some "committed" pairs apart.

Discussion

Substantial experimental support for Fröhlich's theory would impose radical changes on physiological thought. The evidence which Fröhlich recently reviewed (Fröhlich 1980) has made little impact. The

frequency of the predicted oscillations poses technically difficult problems and the results, while indicative of vibrational modes, offer no evidence for the broader implication of the theory. The experiments reported in this paper were designed to test three major aspects of the theory, all of which are necessary features (see Introduction).

The membrane potential

When the pH of the suspending medium is reduced to 6.3 and the chloride determinations indicate that the membrane potential is close to zero, the interaction coefficient is that of a random interaction between cells. In normal cells the interaction coefficient is three times greater and if the pH is restored from 6.3 to 7.6, the interaction returns. In addition to having a membrane potential, erythrocytes have a negative surface charge which keeps cells apart when they are suspended in artificial media devoid of macromolecules. This charge varies with pH and the isoelectric point is in the neighbourhood of pH 2 (Ponder 1948; Bateman and Zellner 1956; Heard and Seaman 1960). The effect, if any, of a reduction to pH 6.3 will therefore be to reduce the surface charge and so act in the opposite direction to the change we have observed. For reasons already given (see Results) an electrostatic repulsion would have a very short range. There was no change in plasma viscosity with change in pH.

Metabolic depletion

Whereas, in the pH experiments, cells which touch proceed to rouleau formation and so make the analysis unequivocal, crenation and sphering of metabolically depleted cells introduce a technical difficulty. Such

cells adhere but they do not form rouleaux and it is a matter of judgement to decide, at the low magnification and resolution perforce used, the instant of contact. When the cellular concentration of ATP falls to about half the normal value, crenation occurs quite abruptly (Nakao 1974) but it is difficult to predict just how long the incubation should last to avoid this transition. In one of our experiments, however, less than 10% of the cells were crenated and the rest were forming rouleaux. The interaction coefficient was close to that in the other three experiments. This obviates the objection that it might be crenation *per se* which led to the low value of Ξ in metabolic depletion.

Nevertheless, the technique may have significant effects in addition to that of cutting off the supply of metabolic energy and so account for the value of $\Xi < 1$ which implies a repulsion, small enough to be overcome by Brownian motion. We know that the membrane potential is halved to 7 mV. There are also ion shifts (Clemens et al. 1980) but the zeta potential and thus the surface charge is unaffected (Weed et al. 1969). Once again there was no change in plasma viscosity.

Metabolic depletion modifies membrane structure in other ways than crenation (Lutz 1978). Indeed, Glansdorff and Prigogine (1971) assert that a supply of metabolic energy is a fundamental thermodynamic necessity, not only for the function of a cellular membrane but also for its stability and for the assembly of its structure.

Glutaraldehyde-fixed cells

Fröhlich's last major requirement is for molecular organization. A great deal is known about the structure of the cellular membrane (Marchesi et al. 1976), the functions of receptors (Edelman 1976), the assembly of proteins (Wickner 1980), and the transport of various substances (Lassen et al. 1980). But of a membrane regarded by Fröhlich as an intricate machine, needing a power supply and undoubtedly subject to control of its function as an entity, we know as yet nothing that enables us to interfere other than destructively. Almost certainly glutaraldehyde, which cross-links proteins and aminophospholipids (Carvalho et al. 1980), wreaks havoc. Some enzyme and receptor activity remains and our method preserves the shape, volume, and area of the cells, but undoubtedly the supply of energy and the membrane potential are lost. So at present we regard this type of experiment as a control along with experiments using microspheres.

Ionophore-treated cells

The treatment of erythrocytes with ionophore A23187 in the manner described caused morphological changes, largely spherizing, in a minority of the cells. This led to difficulties similar to those described for

crenated cells after metabolic depletion. Nevertheless, reduction in the membrane potential by this method reduced the interaction substantially and so confirms the necessity of a membrane potential to the interaction.

General discussion

All the described experiments were done in plasma and there is no detectable attraction between cells suspended in artificial media devoid of macromolecules. Preliminary experiments using PBS containing polyvinylpyrrolidone or Dextran 70, both rouleau-inducing agents (Sewchand and Canham 1979; Sewchand and Bruckschwaiger 1980), suggest that these macromolecules can transmit the interaction. Experiments are in progress to dissect plasma to discover which component is the transmitting medium, but we are as yet unable to suggest what is being transmitted, be they phonons, some other quasiparticle for which there are many candidates, or quantized polar waves. Exchange of phonons between superconducting electrons overcomes electrical repulsion with an attractive force (Fröhlich 1950 cited by Rose-Innes and Rhoderick 1978) so perhaps an analogous process is more than abolishing the Coulomb repulsion between erythrocytes.

The Fröhlich theory predicts that polarized cell membranes may be triggered into coherent, cooperative oscillations if supplied with energy above a minimum rate. The laser is the best known example of such a process but the phonon-maser (Chiao et al. 1964) is the closest comparison we know. The work presented in this paper demonstrates a hitherto unknown interaction of human erythrocytes and we have tested it against Fröhlich's theory. The theory demands a strong transmembrane electric field. When it is abolished experimentally, the interaction disappears, only to reappear when the field is restored. Cutting off ATP also abolishes the interaction but we are in doubt about the integrity of the subtleties of membrane structure as a result, and certainly the electric field is halved. The theory deals with organization, electric field, and energy supply as separate entities but the three are likely to be functionally inseparable. It is an obvious suggestion that the electric field not only polarizes membrane molecules but also is the ultimate supply of energy to the vibrations. The mechanism is there in the work of Mitchell (1976). He uses the term "proticity" to refer to the flow of protons, as electricity refers to the flow of electrons, and describes a number of proticity generators in biological cells. "A generator of proticity plugged through the membrane at any point can act as a source of protonic power than can be drawn upon by a suitable consumer unit plugged through the membrane at any other point, even when the producer and consumer units are quite far apart on a molecular scale . . .

the use of proticity for power transmission in biology is unique in its ubiquitous occurrence and wide range of application." It may therefore be impossible to dissociate, unequivocally, membrane potential difference from the supply of energy.

This interaction of human erythrocytes has features in common with Fröhlich's theoretical predictions. One prediction which is not tested in the present work is that of specificity. The interaction should be attractive, repulsive, or null depending on the relative frequencies or phases of the vibrations of the neighbouring membranes. Evidence of specificity can be deduced from the experiments of Sewchand and Canham (1976) on rouleau formation in mixtures of erythrocytes from different mammalian species. In such interspecies rouleaux, the cells can be recognized by differences in size and they are seen not to be distributed at random within a rouleau. Overbeek (1952) reviewed theoretical and experimental work on the aggregation of polydispersed colloids and showed that the *least* likely collision in Brownian movement is between particles of the *same* size. Yet, in the interspecies experiments, cells of the same size and so of the same species show a preference to adhere to each other. In 10 pairings of species, the only exception was the rat-mouse combination, though the difference in cell size is larger than in some pairs which did show a species preference. As they belong to the same biological family, perhaps all murine erythrocytes are indistinguishable in respect to a Fröhlich interaction. Whether the interspecies interaction is a repulsion, a null interaction, or a smaller attraction than that between cells of the same species cannot be deduced from Sewchand and Canham's (1976) results, but further experiments using methods similar to those described in this paper should answer that question.

It is interesting to note that the history of this line of thought goes back at least 30 years. Fritz London (1950) asking if the study of liquid helium II was of any practical use, wrote "... certain actions between macromolecules in biochemistry could not be understood unless they could be conceived as conditioned by some quantum mechanism involving the system as a whole ...". More recently Glansdorff and Prigogine (1971) wrote "... [There] exist new dynamic states of matter induced by a flow of free energy far from equilibrium ... governed by a new physical chemistry on a supermolecular level, while all the laws referring to the molecular level remain essentially unchanged ...".

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